

COMMAND NEXUS · CIT 140 · WEEK 1

Problem Brief

Meridian Forge Robotics · The Process Below

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SECTION 01

Visible Failure

Nia, the Night QA supervisor reports the MR-17 robot received commands for and began repeating Test K-44 at 2:13 am. Despite an operator’s sent stop request, it would not stop until its current instructions were finished. The current schedule (active scheduler) does not show an any test assigned. Furthermore, Test K-44 was retired months ago and should no longer be running.
(interview) Check the incident log to verify order of events.

SECTION 02

Verified Facts

The current scheduling console shows that no tests are actively assigned to MR-17; not even Test K-44. [current scheduler console]

At 2:13am MR-17 received, accepted, and began a command Test K_44_Start. The operator attempted to cut the test one minute later with a stop request, but it was blocked by a pending active instruction [incident log].

The command came through remote telemetry that was properly formatted from the source label legacy_sync.exe. [Elian Chen, Robotics System Engineer]

SECTION 03

Assumptions and Open Questions

Nia reports that MR-17 was behaving normally before the incident. [interview]
Verify with operating logs.

The maintenance archive records that Test K-44 was removed from the active work flow long before the incident. Was Test K-44 removed from every copy, service, or schedule?

SECTION 04**System Environment Map****SECTION 04A****People and Roles**

Nia Voss (Night QA Supervisor)
Elias Chen (Robotics Systems Engineer)
Mara Ilyan (Production Operations Manager)

SECTION 04B**Processes and Rules**

Active Scheduling Console
MR-17 can receive wireless/remote telemetry commands.
Operators can send commands.

SECTION 04C**Information and Data**

Incident Log
Test Catalog
Operating Schedule
Maintenance Archive

MR-17 returned an ACTIVE_INSTRUCTION_PENDING message when the operator sent a stop request.

SECTION 04D**Technology and Tools**

Robotics Systems Engineer Elias Chen shared that MR-17 received telemetry - transmission of data from a remote source - that was properly formatted for MR-17 and contained the source label legacy_sync.exe. [interview]

Who is the service owner of the telemetry that MR-17 received?

The scheduler console shows no active assignments.

SECTION 04E

Environment and Constraints

The incident was isolated to the QA cell that MR-17 operates in.

SECTION 05

Investigation Boundary

Inside the Investigation

What was the source/owner of the telemetry command that MR-17 received.

Outside for Now

We do not need a restart to clear the current instructions, as the active scheduler does not show any assignments.

SECTION 06

Organizational Impact

Operations Manager Mara reports that production has paused in the affected QA cell. [remote call]. Verify with the operating schedule, production log, and software record.

SECTION 07

Problem Statement

Meridian Forge need to understand why MR-17 received a telemetry command for a retired test that is not in the active scheduler because pausing MR-17 to halt the repeated unscheduled test is forcing production to stop in the QA cell. The investigation must determine the origin of the telemetry command that MR-17 received.

SECTION 08

Next Investigation Step

We first need to verify if Test K-44 removed from every copy, service, or schedule, by checking the operating schedule and software history, and possibly by tracing the origin of the telemetry command.

SHARED ARTIFACT

Working Document Link

https://github.com/atrujillo6801-ai/CIT140__AbrahamTrujillo.git

WEEK 1 STOPPING RULE

Define before designing.

This Problem Brief records the problem that requires investigation. It does not recommend a solution.